

Perturbation Theory:

$$H\Psi = E\Psi = (\lambda^0 H_0 + \lambda^1 H_1 + \lambda^2 H_2 + \dots) (\phi_n + \sum_{k \neq n} \lambda^1 c_{nk}^1 \phi_k + \sum_{k \neq n} \lambda^2 c_{nk}^2 \phi_k + \dots) = (E_n^0 + \lambda E_n^1 + \lambda^2 E_n^2 + \dots) (\phi_n + \sum_{k \neq n} (\lambda c_{nk}^1 + \lambda^2 c_{nk}^2) \phi_k)$$

$$\Rightarrow H_0 |\phi_n\rangle + \sum_{k \neq n} \lambda^1 c_{nk}^1 H_0 |\phi_k\rangle + \sum_{k \neq n} \lambda^2 c_{nk}^2 H_0 |\phi_k\rangle + \lambda H_1 |\phi_n\rangle + \sum_{k \neq n} \lambda^2 c_{nk}^1 H_1 |\phi_k\rangle = E_n^0 |\phi_n\rangle + \sum_{k \neq n} \lambda c_{nk}^1 E_n^0 |\phi_k\rangle + E_n^0 \sum_{k \neq n} \lambda^2 c_{nk}^2 |\phi_k\rangle + \lambda E_n^1 |\phi_n\rangle + \sum_{k \neq n} \lambda^2 c_{nk}^1 E_n^1 |\phi_k\rangle + \lambda^2 E_n^2 |\phi_n\rangle$$

$$\lambda^0 \Rightarrow H_0 |\phi_n\rangle = E_n^0 |\phi_n\rangle \quad \lambda^1 \Rightarrow \sum_{k \neq n} c_{nk}^1 H_0 |\phi_k\rangle + H_1 |\phi_n\rangle = \sum_{k \neq n} c_{nk}^1 E_n^0 |\phi_k\rangle + E_n^1 |\phi_n\rangle$$

$$\lambda^2 \Rightarrow \sum_{k \neq n} c_{nk}^2 H_0 |\phi_k\rangle + \sum_{k \neq n} c_{nk}^1 H_1 |\phi_k\rangle = E_n^0 \sum_{k \neq n} c_{nk}^2 |\phi_k\rangle + \sum_{k \neq n} c_{nk}^1 E_n^1 |\phi_k\rangle + E_n^2 |\phi_n\rangle$$

$$\lambda^0 \Rightarrow H_0 |\phi_n\rangle = E_n^0 |\phi_n\rangle \Rightarrow \langle \phi_n | \Rightarrow \langle \phi_n | H_0 | \phi_n \rangle = E_n^0 \langle \phi_n | \phi_n \rangle //$$

$$\lambda^1 \Rightarrow \sum_{k \neq n} c_{nk}^1 H_0 |\phi_k\rangle + H_1 |\phi_n\rangle = \sum_{k \neq n} c_{nk}^1 E_n^0 |\phi_k\rangle + E_n^1 |\phi_n\rangle$$

$$\Rightarrow \langle \phi_n | \Rightarrow c_{nk}^1 \langle \phi_n | H_0 | \phi_k \rangle + \langle \phi_n | H_1 | \phi_n \rangle = c_{nk}^1 E_n^0 \langle \phi_n | \phi_k \rangle + E_n^1 \langle \phi_n | \phi_n \rangle$$

$$\langle \phi_n | H_1 | \phi_n \rangle = E_n^1 \langle \phi_n | \phi_n \rangle //$$

$$\Rightarrow \langle \phi_m | \Rightarrow c_{nk}^1 \langle \phi_m | H_0 | \phi_k \rangle + \langle \phi_m | H_1 | \phi_n \rangle = c_{nk}^1 E_n^0 \langle \phi_m | \phi_k \rangle + E_n^1 \langle \phi_m | \phi_n \rangle$$

$$c_{nk}^1 \langle \phi_m | H_0 | \phi_k \rangle + \langle \phi_m | H_1 | \phi_n \rangle = c_{nk}^1 E_n^0 \langle \phi_m | \phi_k \rangle + E_n^1 \langle \phi_m | \phi_k \rangle$$

$$c_{nk}^1 E_k^0 \langle \phi_m | \phi_k \rangle + \langle \phi_m | H_1 | \phi_k \rangle = c_{nk}^1 E_n^0 \langle \phi_m | \phi_k \rangle + E_n^1 \langle \phi_m | \phi_k \rangle$$

$$\langle \phi_m | H_1 | \phi_k \rangle = c_{nk}^1 (E_n^0 - E_k^0) \langle \phi_m | \phi_k \rangle + E_n^1 \langle \phi_m | \phi_k \rangle$$

$$\Rightarrow \frac{\langle \phi_m | H_1 | \phi_k \rangle}{E_n^0 - E_k^0} = c_{nk}^1$$

$$\Rightarrow c_{nk}^1 = \frac{\langle \phi_m | H_1 | \phi_n \rangle}{E_n^0 - E_k^0} //$$

$$\langle \phi_n | H_0 | \phi_n \rangle = E_n^0 \langle \phi_n | \phi_n \rangle //$$

$$\langle \phi_n | H_1 | \phi_n \rangle = E_n^1 \langle \phi_n | \phi_n \rangle //$$

$$\lambda^2: \langle n | \quad \lambda^2 \Rightarrow \sum_{k \neq n} c_{nk}^2 H_0 |\phi_k\rangle + \sum_{k \neq n} c_{nk}^1 H_1 |\phi_k\rangle = E_n^0 \sum_{k \neq n} c_{nk}^2 |\phi_k\rangle + \sum_{k \neq n} c_{nk}^1 E_n^1 |\phi_k\rangle + E_n^2 |\phi_n\rangle$$

$$\langle \phi_k | \Rightarrow c_{nk}^2 \langle \phi_k | H_0 | \phi_k \rangle + c_{nk}^1 \langle \phi_k | H_1 | \phi_k \rangle = E_n^0 c_{nk}^2 \langle \phi_k | \phi_k \rangle + c_{nk}^1 E_n^1 \langle \phi_k | \phi_k \rangle + E_n^2 \langle \phi_k | \phi_n \rangle$$

$$\langle \phi_n | \Rightarrow c_{nk}^2 \langle \phi_n | H_0 | \phi_k \rangle + c_{nk}^1 \langle \phi_n | H_1 | \phi_k \rangle = E_n^0 c_{nk}^2 \langle \phi_n | \phi_k \rangle + c_{nk}^1 E_n^1 \langle \phi_n | \phi_k \rangle + E_n^2 \langle \phi_n | \phi_n \rangle$$

$$c_{nk}^1 = \frac{\langle \phi_n | H_1 | \phi_k \rangle}{E_n^0 - E_k^0} \Rightarrow \frac{\langle \phi_n | H_1 | \phi_k \rangle \langle \phi_n | H_1 | \phi_k \rangle}{E_n^0 - E_k^0} = E_n^2$$

$H\Psi = E\Psi = (\lambda^0 H_0 + \lambda H_1) (\phi_n + \sum_{k \neq n} \lambda C_{nk} \phi_k + \lambda^2 C_{nk}^2 \phi_k + \dots) = (E_n^0 + \lambda E_n^1 + \lambda^2 E_n^2 + \dots) (\phi_n + (\sum \lambda C_{nk}^1 + \lambda^2 C_{nk}^2) \phi_k)$
 $\leftarrow$ **Normal Perturbation**

$\rightarrow$ **Degener Perturbation**
 $(\lambda^0 H_0 + \lambda H_1) \Rightarrow (H_0 + \lambda H_1) (\sum_i \alpha_i |\phi_n^i\rangle + \sum_{k \neq n} (\lambda C_{nk}^1 B^i \phi_k + \lambda^2 C_{nk}^2 B^i \phi_k)) = (E_n^0 + \lambda E_n^1 + \lambda^2 E_n^2) (\sum_i \alpha_i \phi_n^i + \sum_{k \neq n} (\lambda C_{nk}^1 B^i \phi_k + \lambda^2 C_{nk}^2 B^i \phi_k))$

$\alpha_i H_0 |\phi_n^i\rangle + \lambda C_{nk}^1 B^i H_0 |\phi_k^i\rangle + \lambda^2 C_{nk}^2 B^i H_0 |\phi_k^i\rangle + \lambda \alpha_i H_1 |\phi_n^i\rangle + \lambda^2 C_{nk}^1 B^i H_1 |\phi_k^i\rangle = E_n^0 \alpha_i |\phi_n^i\rangle + E_n^0 \lambda C_{nk}^1 B^i |\phi_k^i\rangle + E_n^0 \lambda^2 C_{nk}^2 B^i |\phi_k^i\rangle + \lambda \alpha_i E_n^1 |\phi_n^i\rangle + \dots$

$\alpha_i H_0 |\phi_n^i\rangle = E_n^0 \alpha_i |\phi_n^i\rangle$

$C_{nk}^1 B^i H_0 |\phi_k^i\rangle + \alpha_i H_1 |\phi_n^i\rangle = E_n^0 C_{nk}^1 B^i |\phi_k^i\rangle + \alpha_i E_n^1 |\phi_n^i\rangle$

$B^i = B_i \Rightarrow$

$C_{nk}^1 B_i H_1 |\phi_k^i\rangle + \alpha_i H_1 |\phi_n^i\rangle = E_n^0 C_{nk}^1 B_i |\phi_k^i\rangle + \alpha_i E_n^1 |\phi_n^i\rangle$

$\rightarrow$ degenerat als un. $i=1,2$

$\Rightarrow \langle \phi_n^j | \Rightarrow C_{nk}^1 B_i \langle \phi_n^j | H_0 | \phi_k^i \rangle + \alpha_i \langle \phi_n^j | H_1 | \phi_n^i \rangle = E_n^0 C_{nk}^1 B_i \langle \phi_n^j | \phi_k^i \rangle + \alpha_i E_n^1 \langle \phi_n^j | \phi_n^i \rangle$

$\sum_j \alpha_j h_{ji}^{\wedge} = E_n^1 \alpha_i \quad \begin{matrix} j=1 & i=1 & i=2 \\ j=2 & i=1 & i=2 \end{matrix} \Rightarrow$

$\Rightarrow j=1 \Rightarrow \alpha_1 h_{11}^{\wedge} = E_n^1 \alpha_1 \quad \alpha_1 h_{12}^{\wedge} = E_n^1 \alpha_1$
 $j=2 \Rightarrow \alpha_2 h_{21}^{\wedge} = E_n^1 \alpha_2 \quad \alpha_2 h_{22}^{\wedge} = E_n^1 \alpha_2 \Rightarrow \left(\begin{matrix} & \\ & \end{matrix} \right) \left(\begin{matrix} \\ \\ \end{matrix} \right) = 1$

$\begin{pmatrix} h_{11}^{\wedge} & h_{12}^{\wedge} \\ h_{21}^{\wedge} & h_{22}^{\wedge} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = E_n^1 \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$

$\alpha_1 h_{j1} + \alpha_2 h_{j2} = \alpha_j E_n^1 \quad \begin{matrix} \rightarrow \alpha_1 h_{11} + \alpha_2 h_{12} = \alpha_1 E_n^1 \\ \rightarrow \alpha_2 h_{21} + \alpha_2 h_{22} = \alpha_2 E_n^1 \end{matrix}$

